

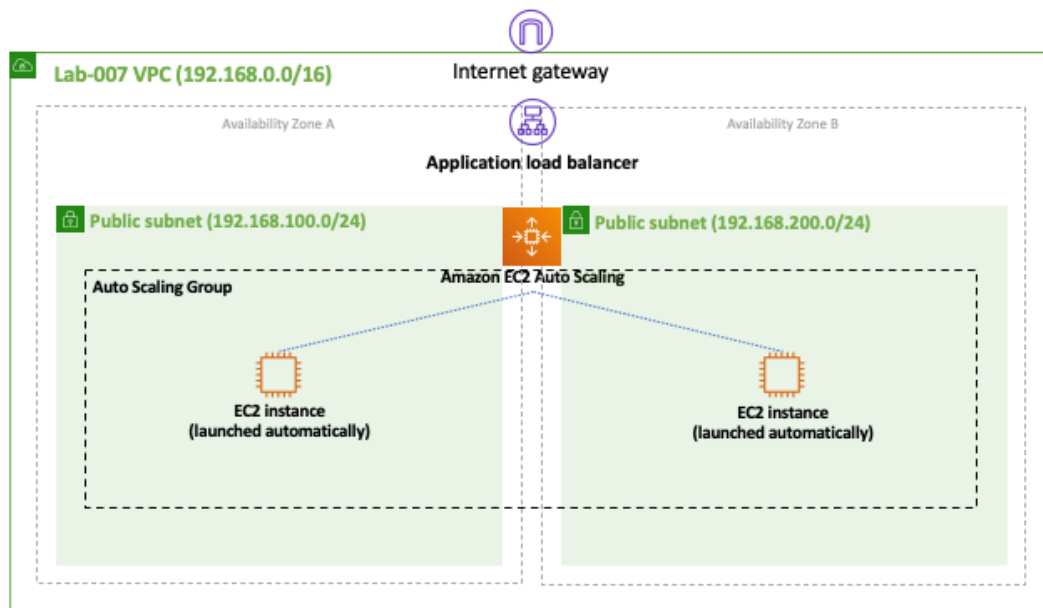
Lab-007

A Simple EC2 Auto Scaling + Load Balancing

Goal

This lab is a continuation of [lab-006](#) with a Load Balancer added.

Architecture Diagram



Overview

Step 1 - Create the Application Load Balancer

Follow [step 5 of lab-005](#), but do not add any targets to the load balancer target group.

Step 2 - Edit the Auto Scaling Group

Edit the auto scaling group adding the load balancer target group created in the previous step.

Create Auto Scaling group

Actions 

Edit

Delete

Filter:

1 to 1 of 1 Auto Scaling Groups

☐	Name	Launch Configuration	Instances	Desired	Min	Max	Availability Zones	Default Cooldown	Health Check Grace
☒	lab-006	lab-006	1	1	1	2	us-west-1a, us-west-1c	300	300

This might be lab-006 or lab-007 depending whether you created this lab after doing lab-006 or not

Edit details - lab-006



Desired Capacity	1
Min	1
Max	2
Availability Zone(s)	us-west-1a us-west-1c
Subnet(s)	subnet-03e6f736175ad48f1(192.168.100.0/24) subnet-A us-west-1a x subnet-08b80bc0195d21ca7(192.168.200.0/24) subnet-B us-west-1c x
Classic Load Balancers	
Target Groups	lab-007 x
Health Check Type	EC2 x
Health Check Grace Period	300
Instance Protection	
Termination Policies	Default x
Suspended Processes	
Max Instance Lifetime	
Placement Groups	
Default Cooldown	300

Consider changing this option to ELB instead of EC2 because ELB health checks also include EC2

Cancel

Save

Test and Validation

Copy the DNS name associated with the load balancer and, using a browser, try to access it. You should be able to see the response associated with the only EC2 instance that is running and automatically launched by the auto scaling service. Now connect to this instance through SSH and run the following commands to artificially increase CPU utilization.

```
stress -c 200
```

You can then begin monitoring the instance's CPU utilization increase and verify that when it reaches 75% or above a new EC2 instance will automatically be launched by the auto scaling service. When that happens, verify that the auto scaling service will automatically launch a second EC2 instance. Now using the DNS associated with the load balancer, verify whether you are able to see responses from both instances as you refresh the page. This test proves that the application load balancer is working properly by redirecting the request evenly to the two instances.